

Every value is $f^*(2\sigma)$ (smallest injected fraction detected at 2σ median significance; lower is better), median over draws. Residual children indented; best test per row **bold**. '>.1' = never reached 2σ .

Galaxy10 (10 classes, frozen ViT, 3 backbones)

backbone / method	Eucl.	Maha.	MMD	SparKer
<i>LeJEPa</i>				
SupCon	.052	.045	.023	.042
SupCon+SIGReg	.027	.028	.029	.028
NPLM-dist	.036	.021	.026	.031
SupCon+cwSIGReg	.043	.068	.036	.052
SupCon $\rightarrow$ res	.054	.037	.040	.044
SupCon $\rightarrow$ res-nplm	.037	.030	.041	.072
SupCon $\rightarrow$ res	.059	.041	.036	.072
<i>VISREG</i>				
SupCon	.033	.054	.029	.031
SupCon+SIGReg	.073	.047	.033	.043
NPLM-dist	.023	.052	.029	.032
SupCon+cwSIGReg	.043	.048	.031	.059
SupCon $\rightarrow$ res	.046	.037	.035	.047
SupCon $\rightarrow$ res-nplm	.035	.052	.033	.031
SupCon $\rightarrow$ res	.073	.052	.038	.062
<i>DINO</i>				
SupCon	.057	.070	.031	.078
SupCon+SIGReg	.069	.071	.029	.079
NPLM-dist	.050	.059	.037	.054
SupCon+cwSIGReg	.062	.068	.039	.087
SupCon $\rightarrow$ res	.069	.080	.033	.097
SupCon $\rightarrow$ res-nplm	.066	.072	.032	.077
SupCon $\rightarrow$ res	.076	.083	.032	.088

Table 1: Galaxy10 pre-discovery (frozen) f^* , median over 5 draws.

backbone / method	Eucl.	Eucl.(a)	Maha.	MMD	SparKer	SpK(a)
<i>LeJEPa</i>						
SupCon	.027	.018	.026	.040	.028	.025
SupCon+SIGReg	.069	.016	.047	.041	.043	.028
NPLM-dist	>.1	.050	.045	.037	>.1	.050
SupCon+cwSIGReg	.065	.030	.038	.051	.046	.044
SupCon $\rightarrow$ res	>.1	.013	.033	.049	.045	.040
SupCon $\rightarrow$ res-nplm	.048	.020	.029	.045	.042	.028
SupCon $\rightarrow$ res	.052	.018	.034	.043	.049	.029
<i>VISREG</i>						
SupCon	.038	.020	.033	.029	.033	.022
SupCon+SIGReg	.048	.017	.043	.036	.034	.025
NPLM-dist	.041	.050	.039	.025	.059	.050
SupCon+cwSIGReg	.045	.020	.050	.027	.048	.047
SupCon $\rightarrow$ res	>.1	.015	.042	.029	.039	.029
SupCon $\rightarrow$ res-nplm	.034	.019	.028	.039	.026	.025
SupCon $\rightarrow$ res	>.1	.020	.034	.033	.047	.034
<i>DINO</i>						
SupCon	.056	.019	.066	.039	.070	.044
SupCon+SIGReg	.062	.021	.076	.034	.077	.047
NPLM-dist	>.1	.030	>.1	.052	.094	.047
SupCon+cwSIGReg	>.1	.020	.056	.084	.078	.045
SupCon $\rightarrow$ res	.078	.021	.070	.066	.080	.044
SupCon $\rightarrow$ res-nplm	.054	.019	.054	.039	.084	.044
SupCon $\rightarrow$ res	>.1	.026	.091	.048	.059	.047

Table 2: Galaxy10 post-discovery f^* , derived cut.

backbone / method	Eucl.	Eucl.(a)	Maha.	MMD	SparKer	SpK(a)
<i>LeJEPa</i>						
SupCon	.026	.012	.027	.038	.039	.024
SupCon+SIGReg	.065	.015	.047	.042	.046	.026
NPLM-dist	.021	.015	.036	.029	.025	.018
SupCon+cwSIGReg	.046	.021	.040	.043	.070	.047
SupCon $\rightarrow$ res	>.1	.014	.033	.042	.046	.039
SupCon $\rightarrow$ res-nplm	.061	.014	.032	.043	.044	.028
SupCon $\rightarrow$ res	.043	.018	.033	.043	.045	.029
<i>VISREG</i>						
SupCon	.038	.017	.033	.028	.027	.028
SupCon+SIGReg	.036	.016	.056	.039	.041	.023
NPLM-dist	.022	.015	.022	.033	.027	.015
SupCon+cwSIGReg	.059	.014	.040	.041	.052	.036
SupCon $\rightarrow$ res	.086	.015	.033	.029	.040	.026
SupCon $\rightarrow$ res-nplm	.037	.019	.035	.037	.027	.025
SupCon $\rightarrow$ res	>.1	.017	.032	.037	.049	.033
<i>DINO</i>						
SupCon	.026	.021	.056	.039	.077	.041
SupCon+SIGReg	.070	.019	.073	.038	.084	.049
NPLM-dist	.059	.026	.079	.041	.071	.043
SupCon+cwSIGReg	.083	.016	.058	.068	.082	.068
SupCon $\rightarrow$ res	.086	.023	.064	.052	.076	.042
SupCon $\rightarrow$ res-nplm	.055	.018	.059	.039	.052	.046
SupCon $\rightarrow$ res	.049	.021	.072	.046	.080	.048

Table 3: Galaxy10 post-discovery f^* , $s/\sqrt{b}$ cut.